

Representation Theory

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Abstract

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1 Basic Definitions of Representations

1.1 Representation of an Algebra

Definition 1.1.1 (Representation of an Algebra)

Let R be a commutative ring. Let A be an R -algebra. A representation of A is a left R -module M and a homomorphism of R -algebras

$$\rho : A \rightarrow \text{End}_R(M)$$

Let A, M be sets. Recall that there is a natural bijection

$$\text{Hom}_{\text{Sets}}(A \times M, M) \cong \text{Hom}_{\text{Sets}}(A, \text{Hom}_{\text{Sets}}(M, M))$$

given by curling. A representation of an R -algebra A is an element in

$$\text{Hom}_{\text{Alg}_R}(A, \text{End}_R(M)) \subseteq \text{Hom}_{\text{Sets}}(A, \text{Hom}_{\text{Sets}}(M, M))$$

Therefore we may ask what kinds of maps of the form $A \times M \rightarrow M$ give a representation of A .

Proposition 1.1.2

Let R be a commutative ring. Let A be an R -algebra. Let M be an R -module. Let $f : A \times M \rightarrow M$ be a function. Then the map $a \mapsto f(a, -)$ defines a representation of A if and only if the following are true.

- f is an R -module homomorphism.
- f defines an A -module structure on M .

The A -module structure can also be seen in the representation $\rho : A \rightarrow \text{End}_R(M)$. It is given by $a \cdot m = \rho(a)(m)$. In particular, not every A -module M gives a representation; the action of A on M must be bilinear in the scalars R . Therefore these modules are often called R -linear A -modules.

However, if $R = k$ is a field and the structure map $k \rightarrow A$ is non-zero, then $k \rightarrow A$ is necessarily injective so that A contains k as a subring. Then the k -linearity of $f : A \times M \rightarrow M$ is null so that A -modules are the same as representations of A over k .

Since representations of an algebra A are A -modules, it follows that the following notions in Rings and Modules apply to representations.

- Subrepresentations.
- Quotient Representations.
- Sum of Representations.

Definition 1.1.3 (Types of Representations)

Let R be a commutative ring. Let A be an R -algebra. Let M be a representation of A .

- We say that M is irreducible if it is simple as an A -module.
- We say that M is semisimple if it is semisimple as an A -module.
- We say that M is indecomposable if it is not completely reducible.

Example 1.1.4

Let k be a field. Define a representation $\rho : k[x] \rightarrow \text{End}_k(k^2) \cong M_2(k)$ by $x \mapsto \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ then extended it R -linearly and respecting products of x . Then ρ is indecomposable but reducible.

Proof

Suppose for a contradiction that ρ is decomposable. Then there exists 1-dimensional linear subspaces L_1, L_2 of k^2 such that ρ_{L_1} and ρ_{L_2} are representations. Suppose that $L_1 = k\langle v_1 \rangle$ and $L_2 = k\langle v_2 \rangle$. Then $x \cdot v_1 \in L_1$ and $x \cdot v_2 \in L_2$ implies that v_1 and v_2 are linearly independent

eigenvectors of $\rho(x)$. But by an easy computation, we see that the only eigenvalue of $\rho(x)$ is 1 and its eigenspace has dimension 1. Hence this is a contradiction.

From the above, we see that the eigenspace of $\rho(x)$ is a subrepresentation of ρ . Hence ρ is reducible. ■

Definition 1.1.5 (The Regular Representation)

Let k be a field. Let A be a k -algebra. Define the regular representation $\text{reg} : A \rightarrow \text{End}_k(A)$ of A by

$$\text{reg}(a)(b) = ab$$

Definition 1.1.6 (Composition Factors)

Let R be a commutative ring. Let A be an R -algebra. Let M be a semisimple representation of A . Suppose that the irreducible decomposition of M is given by

$$M \cong \bigoplus_{i=1}^n M_i$$

Define the composition factors of M to be

$$\text{Comp}(M) = \{M_1, \dots, M_n\}$$

Definition 1.1.7 (Multiplicity of an Irreducible Submodule)

Let R be a commutative ring. Let A be an R -algebra. Let M be a semisimple representation of A . Let N be an irreducible A -module. Define the multiplicity of N in M to be

$$\text{mult}_N(M) = |\{M_i \in \text{Comp}(M) \mid M_i \cong N\}|$$

1.2 Maps between Representations

Definition 1.2.1 (Homomorphisms of Representations)

Let R be a commutative ring. Let A be an R -algebra. Let M, N be representations of A . Let $f : M \rightarrow N$ be a function. We say that f is a homomorphism of representations if the following are true.

- f is an A -module homomorphism.
- f is R -linear.

Since M and N are A -modules, the definition of homomorphism implies that such a morphism must be an A -module homomorphism. Conversely, since not every A -module is an R -linear A -module, A -module homomorphisms are not necessarily morphisms between representations.

Definition 1.2.2 (Set of Homomorphisms of Representations)

Let R be a commutative ring. Let A be an R -algebra. Let M, N be representations of A . Define the set of homomorphisms from M to N to be

$$\text{Hom}_{A_R}(M, N) = \left\{ f : M \rightarrow N \mid \begin{array}{c} f \text{ is a homomorphism} \\ \text{of representations from } M \text{ to } N \end{array} \right\} \subseteq \text{Hom}_A(M, N)$$

Definition 1.2.3 (Isomorphisms of Representations) Let R be a commutative ring. Let A be an R -algebra. Let M, N be R -modules. Let $\rho : A \rightarrow \text{End}_R(M)$ and $\tau : A \rightarrow \text{End}_R(N)$ be representations of A . Let $\phi : M \rightarrow N$ be a homomorphism of representations. We say that ϕ is an isomorphism if there exists a homomorphism of representations $\psi : N \rightarrow M$ such that

$$\psi \circ \phi = \text{id}_M \quad \text{and} \quad \phi \circ \psi = \text{id}_N$$

Lemma 1.2.4

Let R be a commutative ring. Let A be an R -algebra. Let M, N be R -modules. Let $\rho : A \rightarrow \text{End}_R(M)$ and $\tau : A \rightarrow \text{End}_R(N)$ be representations of A . Let $\phi : M \rightarrow N$ be a homomorphism of representations. Then ϕ is an isomorphism if and only if ϕ is an isomorphism of R -modules.

Theorem 1.2.5 (First Isomorphism Theorem for Representations)

Let R be a commutative ring. Let A be an R -algebra. Let M, N be R -modules. Let $\rho : A \rightarrow \text{End}_R(M)$ and $\tau : A \rightarrow \text{End}_R(N)$ be representations of A . Let $\phi : M \rightarrow N$ be a homomorphism of representations. Then the following are true.

- $\ker(\phi)$ is a subrepresentation of M .
- $\text{im}(\phi)$ is a subrepresentation of N .
- There is an isomorphism of representations

$$M / \ker(\phi) \cong \text{im}(\phi)$$

given by ϕ .

Proposition 1.2.6

Let R be a commutative ring. Let A be a finite R -algebra. Let $\rho : A \rightarrow \text{End}_R(M)$ be an irreducible representation of A . Then M is finitely generated.

Proof

Let $0 \neq m \in M$. Then $\rho(A)(m)$ is a submodule of M . Since M is irreducible, we must have $\rho(A)(m) = M$. Since A is finite over R , we must have $\rho(A)(m)$ is finitely generated over R . Hence M is finitely generated over R . ■

1.3 The Set of Representations

Definition 1.3.1 (The Set of Representations)

Let R be a commutative ring. Let A be an R -algebra. Let M be an R -module. Define the space of representations of A on M to be the set

$$\text{Rep}(A, M) = \{[\rho : A \rightarrow \text{End}_R(M)] \mid [\rho] \text{ is an isomorphism class of representations} \}$$

2 Finite Dimensional Representations of Algebra over Algebraically Closed Fields

2.1 Schur's Lemma and its Consequences

Recall from Rings and Modules that we have Schur's lemma: A homomorphism $M \rightarrow N$ of simple R -modules is either the 0 map or an isomorphism.

Lemma 2.1.1 (Schur's Lemma III)

Let k be an algebraically closed field. Let A be a k -algebra. Let V be a finite dimensional k -vector space. Let $\rho : A \rightarrow \text{End}_k(V)$ be an irreducible representations of A . Let $\phi : V \rightarrow V$ be a homomorphism of representations. Then $\phi = \lambda \text{id}_V$ for some $\lambda \in k$.

Proof

Since ϕ is a homomorphism of simple A -modules, by Schur's lemma we know that $\phi = 0$ or ϕ is an isomorphism. If $\phi = 0$ then we are done. Otherwise, since k is algebraically closed, the endomorphism ϕ admits an eigenvalue $\lambda \in k$ and eigenvector $v \in V$. Consider the homomorphism of representations $\phi - \lambda \text{id}_V$. By Schur's lemma, this map is either 0 or an isomorphism. But it is not an isomorphism since $(\phi - \lambda \text{id}_V)(v) = 0$. Hence $\phi - \lambda \text{id}_V = 0$. So we are done. ■

Corollary 2.1.2

Let k be an algebraically closed field. Let A be a commutative k -algebra. Let V be a finite dimensional k -vector space. Let $\rho : A \rightarrow \text{End}_k(V)$ be an irreducible representations of A . Then $\dim(V) = 1$.

Proof

By Schur's lemma III, we know that $\rho(a) : V \rightarrow V$ is a scalar operator for all $a \in A$. This means that $\rho(a)(v) = \lambda_a v$ for some $\lambda_a \in k$ and all $v \in V$. For any fixed $v_0 \in V$, this implies that $k\langle v_0 \rangle$ is a subrepresentation of ρ . If $\dim(V) \geq 2$, then $k\langle v_0 \rangle$ is a non-trivial subrepresentation, contradicting the irreducible assumption. Hence $\dim(V) = 1$. ■

In other words, every finite dimensional representation of a commutative algebra over an algebraically closed field must be of dimension 1. And these are easy to classify.

Proposition 2.1.3

Let k be an algebraically closed field. Let A be a k -algebra. Let V, W be finite dimensional irreducible representations of A . Then we have

$$\dim_k(\text{Hom}_{A_k}(V, W)) = \begin{cases} 1 & \text{if } V \cong W \\ 0 & \text{otherwise} \end{cases}$$

Proof

If V is not isomorphic to W , then by Schur's lemma we must have $\text{Hom}_{\text{Rep}(A)}(V, W) \subseteq \text{Hom}_A(V, W) = \{0\}$. If $V = W$, then Schur's lemma III implies that id_V generates the k -vector space. ■

Proposition 2.1.4

Let k be an algebraically closed field. Let A be a k -algebra. Let V_i, W_j be finite dimensional irreducible representations of A for $1 \leq i \leq n$ and $1 \leq j \leq m$. Then we have

$$\dim_k \left(\text{Hom}_{A_k} \left(\bigoplus_{i=1}^n V_i, \bigoplus_{j=1}^m W_j \right) \right) = \sum_{i=1}^n \sum_{j=1}^m \dim_k(\text{Hom}_{A_k}(V_i, W_j))$$

Corollary 2.1.5

Let k be an algebraically closed field. Let A be a k -algebra. Let V, W be finite dimensional representations of A such that W is irreducible. Then we have

$$\dim_k(\operatorname{Hom}_{A_k}(V, W)) = \operatorname{mult}_W(V)$$

2.2 Character Theory

Let k be an algebraically closed field. Recall that the trace of a k -linear map between finite dimensional vector spaces have a basis independent description.

Definition 2.2.1 (Characters of a Representation)

Let k be an algebraically closed field. Let A be a k -algebra. Let $\rho : A \rightarrow \operatorname{End}_k(V)$ be a finite dimensional representation of A . Define the character of ρ to be function $\chi_V : A \rightarrow k$ given by

$$\chi_V(a) = \operatorname{tr}(\rho(a))$$

Proposition 2.2.2

Let k be an algebraically closed field. Let A be a k -algebra. Let V, W be isomorphic finite dimensional representations of A . Then $\chi_V = \chi_W$.

Proposition 2.2.3

Let k be an algebraically closed field. Let A be a k -algebra. Let $\rho : A \rightarrow \operatorname{End}_k(V)$ be a finite dimensional representation of A . Then χ_V descends to a well defined function $\chi_V : \frac{A}{[A, A]} \rightarrow k$.

Lemma 2.2.4

Let k be an algebraically closed field. Let A be a k -algebra. Let V be a finite dimensional representation of A . Let $W \leq V$ be a subrepresentation. Then we have

$$\chi_V = \chi_W + \chi_{V/W}$$

Definition 2.2.5 (Irreducible Characters)

Let k be an algebraically closed field. Let A be a k -algebra. Let V be a representation of A . We say that χ_V is irreducible if V is irreducible.

Proposition 2.2.6

Let k be an algebraically closed field. Let A be a k -algebra. Let $V_1, \dots, V_n$ be finite dimensional irreducible representations of A . Then $\{\chi_{V_1}, \dots, \chi_{V_n}\}$ are linearly independent in $\operatorname{Hom}_k(A/[A, A], k)$.

Definition 2.2.7 (Characters of an Algebra)

Let k be an algebraically closed field. Let A be a k -algebra. Let $\phi : A \rightarrow k$ be a function. We say that ϕ is a character of A if there exists a finite dimensional representation V of A such that $\phi = \chi_V$.

2.3 The Semisimple Condition**Proposition 2.3.1**

Let k be an algebraically closed field. Let A be a finite semisimple k -algebra. Let $V_1, \dots, V_n$ be the complete list of non-isomorphic representations of A . Then there is a k -algebra isomorphism

$$A \cong \bigoplus_{i=1}^n \operatorname{End}_k(V_i)$$

Proof

By the Artin Wedderburn theorem, there is a k -algebra isomorphism

$$A \cong \bigoplus_{i=1}^p M_{n_i}(D_i)$$

where each D_i is a division algebra over k . Since k is algebraically closed, we have $D_i = k$. By a corollary of the Artin-Wedderburn theorem, the complete list of simple A -modules is given by the unique simple $M_{n_i}(k)$ -submodule T_i of itself. Hence $n = p$. Moreover, by reordering the index, we have isomorphisms of $M_{n_i}(k)$ -modules $V_i \cong T_i$ that are isomorphisms of A -modules (since the ring structure of the product of matrices is given by the product of the ring structures of the matrix ring, and the Artin-Wedderburn isomorphism is a ring isomorphism). Now $T_i \cong k^{n_i}$ as k -vector spaces. This gives an isomorphism $V_i \cong k^{n_i}$ of k -vector spaces and hence an isomorphism of rings $\text{End}_k(V_i) \cong M_{n_i}(k)$. Thus we have

$$A \cong \bigoplus_{i=1}^p M_{n_i}(k) \cong \bigoplus_{i=1}^n \text{End}_k(V_i)$$

■

Corollary 2.3.2

Let k be an algebraically closed field. Let A be a finite semisimple k -algebra. Let $V_1, \dots, V_n$ be the complete list of non-isomorphic representations of A . Then we have

$$\dim_k(A) = \sum_{i=1}^n \dim_k(V_i)^2$$

Corollary 2.3.3

Let k be an algebraically closed field. Let A be a finite k -algebra. Let V be an irreducible representation of A . Then we have

$$\text{mult}_V(A) = \dim_k(\text{Hom}_{A_k}(A, V)) = \dim_k(V)$$

Proof

Direct from Artin-Wedderburn theorem and its corollaries.

■

Proposition 2.3.4

Let k be an algebraically closed field. Let A be a finite semisimple k -algebra. Let $V_1, \dots, V_n$ be the complete list of non-isomorphic irreducible finite dimensional representation of A . Then $\{\chi_{V_1}, \dots, \chi_{V_n}\}$ is a basis for $\text{Hom}_k(A/[A, A], k)$.

Corollary 2.3.5

Let k be an algebraically closed field. Let A be a finite semisimple k -algebra. Let χ be a character of A . Then there exists a representation V of A such that $\chi = \chi_V$.

3 Representation Theory of Finite Groups

3.1 Structural Results

By the natural isomorphism

$$\mathrm{Hom}_{\mathbf{Alg}_k}(k[G], \mathrm{End}_k(V)) \cong \mathrm{Hom}_{\mathbf{Grp}}(G, \mathrm{End}_k(V))$$

we see that representations of a group is the same as representation of the group ring.

Example 3.1.1

The following are representations of $D_8 = \langle r, s \mid r^4, s^2, rsrs \rangle$.

- $\rho : D_8 \rightarrow GL_2(\mathbb{C})$ defined by $\rho(r) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ and $\rho(s) = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$.
- $\rho : D_8 \rightarrow GL_n(\mathbb{C})$ defined by $\rho(r) = \rho(s) = I_n$.
- $\rho : D_8 \rightarrow \mathbb{C}^\times$ defined by $\rho(r) = \pm 1$ and $\rho(s) = \pm 1$.
- $\rho : D_8 \rightarrow GL_4(\mathbb{C})$ defined by $\rho(r) = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}$ and $\rho(s) = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$.

Definition 3.1.2 (Permutation Representations)

Let X be a set. Let G be a group acting on X . Let k be a field. Define the permutation representation $\rho : G \rightarrow GL(k\langle X \rangle)$ of G on X over k by the formula

$$\rho(g)(x) = g \cdot x$$

and then extending it k -linearly.

Every group G has an action on itself by left multiplication. In this case the permutation representation is just the regular representation.

Example 3.1.3

The representation of $C_3 = \langle x \mid x^3 \rangle$ on $\mathbb{F}_3^2$ given by

$$\rho(x) = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

is not semisimple.

Let k be an algebraically closed field. Let G be a finite group. Suppose that $\mathrm{char}(k)$ does not divide $|G|$. Maschke's theorem which implies that $k[G]$ is semisimple is a powerful tool to classify irreducible representations of G because of the corollaries it gives from section 1.

- Since $|G|$ is finite, every representation of G is finite dimensional. Moreover, there are only finitely many non-isomorphic irreducible representations $V_1, \dots, V_n$.
- Combining the above corollary, we have a decomposition of the regular representation into irreducible representations

$$k[G] \cong \bigoplus_{i=1}^n \bigoplus_{j=1}^{\dim(V_i)} V_i$$

- This gives the formula

$$|G| = \sum_{i=1}^n \dim_k(V_i)^2$$

Example 3.1.4

Let G be a group of order 8. Let k be an algebraically closed field such that $\mathrm{char}(k) \neq 2$. Let

$V_1, \dots, V_n$ be the complete list of non-isomorphic irreducible $k[G]$ -modules. Then exactly one of the following is true (up to reordering of the modules).

- $n = 8$ and $\dim(V_i) = 1$.
- $n = 5$ and $\dim(V_1) = \dots = \dim(V_4) = 1$ and $\dim(V_5) = 2$.

Similar to the case of representations, the character of a representation of a finite group $\chi : k[G] \rightarrow k$ is the same information as a function $\chi : G \rightarrow k$ simply by restriction and extension via scalars.

Proposition 3.1.5

Let G be a finite group. Let k be an algebraically closed field such that $\text{char}(k)$ does not divide $|G|$. Let $\rho : G \rightarrow \text{End}_k(V)$ be a representation of G . Then the following are true.

- $\chi_V(1_G) = \dim(V)$.
- $\chi_V(hgh^{-1}) = \chi(g)$ for all $g, h \in G$.

Proof

By Linear Algebra, we have that

$$\chi_V(1_G) = \text{tr}(\rho(1_G)) = \text{tr}(\text{id}_V) = \text{tr}(I_{\dim(V)}) = \dim(V)$$

so $\chi_V(1_G) = \dim(V)$.

Let $g, h \in G$. Then we have

$$\chi_V(hgh^{-1}) = \text{tr}(\rho(h)\rho(g)\rho(h^{-1})) = \text{tr}(\rho(h)\rho(h^{-1})\rho(g)) = \text{tr}(\rho(hh^{-1})\rho(g)) = \text{tr}(\rho(g)) = \chi_V(g)$$

by properties of the trace map. ■

The second item implies that χ descends to a function on conjugation classes $\chi : \text{Cl}(G) \rightarrow k$.

Proposition 3.1.6

Let G be a finite group. Let $\rho : G \rightarrow \text{End}_{\mathbb{C}}(V)$ be a complex representation of G . Let $g \in G$. Then the following are true.

- Let ω be a $|g|$ th root of unity. Then we have

$$\chi(g) = \sum_{i=1}^k \omega^{j_i}$$

for some $j_i \in \mathbb{N}$ and $k \in \mathbb{N}$.

- We have $\chi_V(g^{-1}) = \overline{\chi_V(g)}$.

Proof

Let $H = \langle g \rangle$. Then V is also a $\mathbb{C}[H]$ -module. Let $V = \bigoplus_{i=1}^m V_i$ be the decomposition of V into irreducible $\mathbb{C}[H]$ -submodules. Since H is cyclic, we must have $\dim(V_i) = 1$ by cor2.1.2. Choose non-zero $v_i \in V_i$. Then $E = \{v_1, \dots, v_m\}$ is a basis for V and $[\rho(g)]_E^E$ is a diagonal matrix. Now for each i , V_i is a $\mathbb{C}[H]$ -module. Hence $\rho(g)(v_i) \in V_i$ implies that $\rho(g)(v_i) = \lambda_i v_i$ for some $\lambda_i \in \mathbb{C}^\times$. Then we have $\lambda^{j_i} v_i = \rho(g)^{|g|}(v_i) = v_i$ so that λ is a $|g|$ th root of unity. Since $\rho(g)$ is diagonal, we conclude that the trace of $\rho(g)$ is a sum of $|g|$ th roots of unity.

From the above proof, we know that $[\rho(g)]_E^E = \text{diag}(\omega^{-j_1}, \dots, \omega^{-j_m}) = \text{diag}(\overline{\omega^{j_1}}, \dots, \overline{\omega^{j_m}})$. Hence $\chi_V(g^{-1}) = \overline{\chi_V(g)}$. ■

3.2 The Dual and Tensor Products of Representations

Definition 3.2.1 (The Dual Representation)

Let G be a finite group. Let $\rho : G \rightarrow \text{End}_{\mathbb{C}}(V)$ be a complex representation of G . Define the dual of ρ to be the representation $\rho^* : G \rightarrow \text{End}_{\mathbb{C}}(V^*)$ given by

$$\rho^*(g) = \rho(g^{-1})^T$$

Lemma 3.2.2

Let G be a finite group. Let $\rho : G \rightarrow \text{End}_{\mathbb{C}}(V)$ be a complex representation of G . Then we have $\rho^*(g)(f) = f \circ \rho(g)$ for $f \in V^*$.

Proposition 3.2.3

Let G be a finite group. Let V be a $\mathbb{C}[G]$ -module. Then the following are true.

- We have $\chi_{V^*} = \overline{\chi_V}$.
- V is irreducible if and only if V^* is irreducible.

3.3 Orthogonality of Characters of Irreducible Representations

Let G be a finite group. Since $\mathbb{C}[G]$ is a semisimple $\mathbb{C}$ -algebra, we know that there are $\mathbb{C}$ -vector space isomorphisms

$$\begin{aligned} \text{Hom}_{\mathbb{C}}\left(\frac{\mathbb{C}[G]}{[\mathbb{C}[G], \mathbb{C}[G]]}, \mathbb{C}\right) &= \{f \in \text{Hom}_{\mathbb{C}}(\mathbb{C}[G], \mathbb{C}) \mid f(ghg^{-1}h^{-1}) = 0 \text{ for all } g, h \in G\} \\ &= \{f \in \text{Hom}_{\text{Set}}(G, \mathbb{C}) \mid f(ghg^{-1}) = f(h) \text{ for all } g, h \in G\} \\ &= \text{Hom}_{\text{Set}}(\text{Cl}(G), \mathbb{C}) \end{aligned}$$

By a previous result, the vector space has a basis given by the characters of G . We now define an inner product on the $\mathbb{C}$ -space.

Definition 3.3.1 (The Inner Product Space of Class Functions)

Let G be a finite group. Define the inner product space of class functions of G to be the $\mathbb{C}$ -vector space

$$\text{Hom}_{\text{Set}}(\text{Cl}(G), \mathbb{C})$$

together with the inner product given by

$$\langle f_1, f_2 \rangle = \frac{1}{|G|} \sum_{g \in G} f_1(g) \overline{f_2(g)}$$

for $f_1, f_2 \in \text{Hom}_{\text{Set}}(\text{Cl}(G), \mathbb{C})$.

Proposition 3.3.2

Let G be a finite group. Let ϕ, ψ be characters of G . Then we have $\langle \phi, \psi \rangle \in \mathbb{R}$.

Proof

Let ϕ, ψ be characters. Then we have

$$\begin{aligned}
 \overline{\langle \phi, \psi \rangle} &= \overline{\frac{1}{|G|} \sum_{g \in G} \phi(g) \psi(g)} \\
 &= \frac{1}{|G|} \sum_{g \in G} \overline{\phi(g)} \psi(g) \\
 &= \frac{1}{|G|} \sum_{g \in G} \phi(g^{-1}) \psi(g) \\
 &= \frac{1}{|G|} \sum_{h \in G} \phi(h) \psi(h^{-1}) \quad (\text{Let } h = g^{-1}) \\
 &= \langle \phi, \psi \rangle
 \end{aligned}$$

so we are done. ■

Lemma 3.3.3

Let G be a finite group. Let V, W be irreducible $\mathbb{C}[G]$ -modules whose representation is given by ρ and σ respectively. Let $f : V \rightarrow W$ be a morphism. Let $\tilde{f} : V \rightarrow W$ be the map defined by

$$\tilde{f}(v) = \frac{1}{|G|} \sum_{g \in G} \rho(g^{-1}) \circ f \circ \sigma(g)$$

Then $\tilde{f}$ is a $\mathbb{C}[G]$ -module homomorphism. Moreover, we have

$$\tilde{f} = \begin{cases} \frac{\text{tr}(f)}{\dim(V)} \text{id}_V & \text{if } V \cong W \\ 0 & \text{otherwise} \end{cases}$$

Proposition 3.3.4

Let G be a finite group. Let V, W be irreducible $\mathbb{C}[G]$ -modules. Then we have

$$\langle \chi_V, \chi_W \rangle = \begin{cases} 1 & \text{if } V \cong W \\ 0 & \text{otherwise} \end{cases}$$

Theorem 3.3.5

Let G be a finite group. Let V, W be $\mathbb{C}[G]$ -modules such that W is irreducible. Then we have

$$\langle \chi_V, \chi_W \rangle = \text{mult}_W(V)$$

Corollary 3.3.6

Let G be a finite group. Let V, W be irreducible $\mathbb{C}[G]$ -modules. Then $V \cong W$ if and only if $\chi_V = \chi_W$.

Proposition 3.3.7

Let G be a finite group. Let V be a $\mathbb{C}[G]$ -module. Then V is irreducible if and only if $\langle \chi_V, \chi_V \rangle = 1$.

Lemma 3.3.8

Let G be a finite group. Then we have

$$\dim_{\mathbb{C}}(\text{Hom}_{\text{Set}}(\text{Cl}(G), \mathbb{C})) = |\text{Cl}(G)|$$

Lemma 3.3.9

Let G be a finite group. Let $\rho : G \rightarrow \text{End}_{\mathbb{C}}(V)$ be an irreducible representation of G . Let $f \in \text{Hom}_{\text{Set}}(\text{Cl}(G), \mathbb{C})$. Let $\varphi : V \rightarrow V$ be the map defined by

$$\varphi(v) = \sum_{g \in G} f(g) \rho(g)v$$

Then we have

$$\varphi(v) = \frac{|G|}{\dim(V)} \langle f, \overline{\chi_V} \rangle v$$

for all $v \in V$.

Lemma 3.3.10

Let G be a finite group. Let $f \in \text{Hom}_{\text{Set}}(\text{Cl}(G), \mathbb{C})$. Let $V_1, \dots, V_n$ be the complete list of non-isomorphic irreducible $\mathbb{C}[G]$ -modules. If $\langle f, \chi_{V_i} \rangle$ for all $1 \leq i \leq n$, then we have $f = 0$.

Proposition 3.3.11

Let G be a finite group. Let $V_1, \dots, V_n$ be a complete list of non-isomorphic irreducible $\mathbb{C}[G]$ -modules. Then $\{\chi_{V_1}, \dots, \chi_{V_n}\}$ is an orthonormal basis for $\text{Hom}_{\text{Set}}(\text{Cl}(G), \mathbb{C})$.

Corollary 3.3.12

Let G be a finite group. Then there are exactly $|\text{Cl}(G)|$ non-isomorphic irreducible $\mathbb{C}[G]$ -modules.

Corollary 3.3.13

Let G be a finite group. Let $g, h \in G$. Then g and h are conjugate if and only if $\chi(g) = \chi(h)$ for all characters χ of G .

Corollary 3.3.14

Let G be a finite group. Let $g \in G$. Then g and g^{-1} are conjugate if and only if $\chi(g) \in \mathbb{R}$ for all characters χ of G .

3.4 Character Tables

Definition 3.4.1 (The Character Table of a Finite Group)

Let G be a finite group. Let $V_1, \dots, V_{|\text{Cl}(G)|}$ be the complete list of non-isomorphic irreducible $\mathbb{C}[G]$ -modules. Let $\text{Cl}(g_1), \dots, \text{Cl}(g_{|\text{Cl}(G)|})$ be the complete list of distinct conjugacy classes of G . Define the character table of G to be the matrix $(a_{i,j}) \in M_{|\text{Cl}(G)|}(\mathbb{C})$ defined by

$$a_{i,j} = \chi_{V_i}(g_j)$$

Example 3.4.2

The character table for $\mathbb{Z}/3\mathbb{Z}$ is given by

$\mathbb{Z}/3\mathbb{Z}$	[0]	[1]	[2]
χ_{Trivial}	1	1	1
χ_2	1	$e^{2\pi i/3}$	$e^{4\pi i/3}$
χ_3	1	$e^{4\pi i/3}$	$e^{2\pi i/3}$

where χ_2 is the character associated to the representation $\mathbb{Z}/3\mathbb{Z} \rightarrow \mathbb{C}$ given by $[1] \mapsto e^{2\pi i/3}$ and χ_3 is the character associated to the representation $\mathbb{Z}/3\mathbb{Z} \rightarrow \mathbb{C}$ given by $[1] \mapsto e^{4\pi i/3}$.

Example 3.4.3

The character table for $D_6 = \langle r, s \mid r^3, s^2, rsrs \rangle$ is given by

D_6	1	r	s
χ_{Trivial}	1	1	1
χ_2	1	1	-1
χ_3	2	-1	0

where χ_2 is the character associated to the representation $D_6 \rightarrow \mathbb{C}$ given by $r \mapsto 1, s \mapsto -1$ and χ_3 is the character associated to the representation

Example 3.4.4

The character table for A_4 is given by

A_4	1	(1 2) (3 4)	(1 2 3)	(1 3 2)
χ_{Trivial}	1	1	1	1
χ_2	1	1	$e^{2\pi i/3}$	$e^{4\pi i/3}$
χ_3	1	1	$e^{4\pi i/3}$	$e^{2\pi i/3}$
χ_3	3	-1	0	0

where χ_2 is the character associated to the representation $D_6 \rightarrow \mathbb{C}$ given by $r \mapsto 1, s \mapsto -1$ and χ_3 is the character associated to the representation

Proposition 3.4.5

Let G be a finite group. Let $A = (a_{i,j}) \in M_{|\text{Cl}(G)|}(\mathbb{C})$ be the character table of G . Then the following are true.

- The rows of A are orthogonal:

$$\sum_{j=1}^{|\text{Cl}(G)|} \frac{a_{r,j} \overline{a_{s,j}}}{|C_G(g_j)|} = \delta_{r,s}$$

- The columns of A are orthogonal:

$$\sum_{i=1}^{|\text{Cl}(G)|} \frac{a_{i,r} \overline{a_{i,s}}}{|C_G(g_s)|} = \delta_{r,s}$$

Motivated by the above proposition, we define an augmented version of the character table as follows.

Definition 3.4.6 (The Augmented Character Table)

Let G be a finite group. Let $V_1, \dots, V_{|\text{Cl}(G)|}$ be the complete list of non-isomorphic irreducible $\mathbb{C}[G]$ -modules. Let $\text{Cl}(g_1), \dots, \text{Cl}(g_{|\text{Cl}(G)|})$ be the complete list of distinct conjugacy classes of G . Define the character table of G to be the matrix $(a_{i,j}) \in M_{|\text{Cl}(G)|}(\mathbb{C})$ defined by

$$a_{i,j} = \frac{\chi_{V_i}(g_j)}{|C_G(g_j)|}$$

Corollary 3.4.7

Let G be a finite group. Let A be the augmented character table of G . Then A is an orthonormal matrix.

A number of previous theorems help determining the character table. To compute the number of conjugacy classes, we may use the following facts. Suppose that $V_1, \dots, V_n$ is the complete list of non-isomorphic irreducible representations of a finite group G .

- Use the fact that $|G| = \sum_{i=1}^n \dim(V_i)^2$ and arithmetic to figure out n .
- $|\text{Cl}(G)| = n$.

To then figure out the character table, we may use the following facts.

- $\chi_{V_i}(1_G) = \dim(V_i)$.
- The trivial map $G \rightarrow \mathbb{C}$ is always a representation of G .

- Orthogonality relations between the rows and the columns
- $\chi(g^{-1}) = \overline{\chi(g)}$ for all characters χ .
- g and g^{-1} are conjugate if and only if $\chi(g) \in \mathbb{R}$ for all characters χ .
- If χ is an irreducible character, then $\overline{\chi}$ is also an irreducible character.

Proposition 3.4.8

Let G be a finite group. Let χ be a character of G . Let $\rho : G \rightarrow \mathbb{C}^\times$ be a representation of G . Then the following are true.

- $\chi\rho$ is a representation of G .
- If χ is irreducible, then $\chi\rho$ is irreducible.

3.5 Representation Theory of Subgroups

Definition 3.5.1 (Restriction of Representation to a Subgroup)

Let k be a field. Let G be a group. Let $H \leq G$ be a subgroup. Let V be a representation of G over k .

- Define the restriction of V to H to be the vector space

$$\text{Res}_H^G(V) = V$$

together with the action of $k[H]$ given by restricting the action of $k[G]$ to $k[H]$.

- Define the restriction of χ_V to H by $\text{Res}_H^G \chi_V$.

Note: Even if V is an irreducible $k[G]$ -module, it may not be an irreducible $k[H]$ -module.

Proposition 3.5.2

Let G be a finite group. Let $H \leq G$ be a subgroup. Let ψ be a non-zero character of H . Then there exists an irreducible character χ of G such that $\langle \text{Res}_H^G \chi, \psi \rangle \neq 0$.

Proof

Let $\chi_1, \dots, \chi_k$ be the complete list of irreducible characters of G . Let χ_{reg} be the character of the regular representation of G . Then we know that $\chi_{\text{reg}} = \sum_{i=1}^k \chi_i(1_G) \chi_i$. Notice that the regular $\mathbb{C}[G]$ -module contains $\mathbb{C}[H]$ and $\mathbb{C}[H]$ is closed under the action of H . Hence $\mathbb{C}[H]$ is a submodule of $\text{Res}_H^G(\mathbb{C}[G])$.

Suppose that ψ is irreducible. Then the corresponding $\mathbb{C}[H]$ -module V is a submodule of the regular representation $\mathbb{C}[H]$. Thus $\langle \text{Res}_H^G \chi_{\text{reg}}, \psi \rangle \neq 0$. Then we have

$$0 \neq \left\langle \sum_{i=1}^k \chi_i(1_G) \text{Res}_H^G \chi_i, \psi \right\rangle = \sum_{i=1}^k \chi_i(1_G) \langle \text{Res}_H^G \chi_i, \psi \rangle$$

Thus there exists χ_i such that $\langle \text{Res}_H^G \chi_i, \psi \rangle \neq 0$.

We can extend this to the case where ψ is reducible by noting that for a character ψ_j for some irreducible composition factor of ψ , we have $\langle \text{Res}_H^G \chi_i, \psi_j \rangle \neq 0$. ■

Theorem 3.5.3 (Clifford's Theorem)

Let G be a finite group. Let $H \trianglelefteq G$ be a normal subgroup. Let χ be an irreducible character of G . Let $\psi_1, \dots, \psi_n$ be the complete list of irreducible characters of H such that $\text{Res}_H^G \chi$ is a linear combination of $\psi_1, \dots, \psi_n$ with non-zero coefficients. Then the following are true.

- Each ψ_i have the same dimension.
- $\langle \text{Res}_H^G \chi, \psi_i \rangle$ is the same for all i .

Proof

Let V be an irreducible $\mathbb{C}[G]$ -module corresponding to χ . Let U be an irreducible $\mathbb{C}[H]$ -module of $\text{Res}_H^G(V)$. For all $g \in G$, the set $g \cdot U$ is a subspace of V . For all $h \in H$, we have

$$h(gu) = g(g^{-1}hg)u \in gU$$

since U is a $\mathbb{C}[H]$ -module and H is normal implies $g^{-1}hg \in H$. So $g \cdot U$ is a $\mathbb{C}[H]$ -submodule of $\text{Res}_H^G(V)$.

I claim that $g \cdot U$ is irreducible. Suppose that W is a $\mathbb{C}[H]$ -submodule of $g \cdot U$. Then $g^{-1} \cdot W$ is a $\mathbb{C}[H]$ -submodule of U by the above paragraph. Hence $g^{-1} \cdot W$ is U or 0 since U is irreducible. Hence W is equal to $g \cdot U$ or 0 . Thus $g \cdot U$ is irreducible. Since $g \cdot -$ is an invertible linear map, we have $\dim_{\mathbb{C}}(g \cdot U) = \dim_{\mathbb{C}}(U)$.

Let $g_1, \dots, g_l$ be complete list of distinct elements in G such that $g_1 \cdot U, \dots, g_l \cdot U$ are pairwise not equal to each other. Then $\bigoplus_{j=1}^l g_j \cdot U$ is a $\mathbb{C}[G]$ -submodule of V . Since V is an irreducible $\mathbb{C}[G]$ -module, we have $\bigoplus_{g \in G} g \cdot U = V$. Hence V is the direct sum of these irreducible submodules of equal dimension. This proves part 1 of the theorem. ■

Proposition 3.5.4

Let G be a finite group. Let $H \leq G$ be a subgroup. Let χ be an irreducible character of G . Let $\psi_1, \dots, \psi_r$ be the complete list of irreducible characters of H . Suppose that $\text{Res}_H^G \chi = \sum_{i=1}^r d_i \psi_i$. Then the following are true.

- We have $\sum_{i=1}^r d_i^2 \leq [G : H]$.
- The above equality holds if and only if $\chi(g) = 0$ for all $g \in G \setminus H$.

Proof

Notice that

$$\sum_{i=1}^r d_i^2 = \langle \text{Res}_H^G \chi, \text{Res}_H^G \chi \rangle_H = \frac{1}{|H|} \sum_{h \in H} \chi(h) \overline{\chi(h)}$$

and also

$$1 = \langle \chi, \chi \rangle_G = \frac{1}{|G|} \left(\sum_{h \in H} \chi(h) \overline{\chi(h)} + \sum_{g \in G \setminus H} \chi(g) \overline{\chi(g)} \right) = \frac{|H|}{|G|} \left(\sum_{i=1}^r d_i^2 \right) + \frac{1}{|G|} \sum_{g \in G \setminus H} \chi(g) \overline{\chi(g)}$$

so we are done. ■

Definition 3.5.5 (The Induced Representation)

Let G be a finite group. Let $H \leq G$ be a subgroup. Let $g_1 H, \dots, g_n H$ be the complete list of distinct left cosets of H in G . Let U be a $\mathbb{C}[H]$ -module.

- Define the induced representation of U to be

$$\text{Ind}_H^G(U) = \bigoplus_{i=1}^n g_i U$$

together with the action of G given by left multiplication.

- Denote the character of the induced representation of χ by $\text{Ind}_H^G \chi$.

Example 3.5.6

Let $G = D_3 = \langle a, b \mid a^3, b^2, abab \rangle = S_3$. Let $H = \langle a \rangle \leq D_3$. Then the complete list of irreducible $\mathbb{C}[H]$ -modules are given by

- $W_1 = \mathbb{C}\langle 1_H + a + a^2 \rangle$.
- $W_2 = \mathbb{C}\langle 1_H + \omega a + \omega^2 a^2 \rangle$.

- $W_3 = \mathbb{C}\langle 1_H + \omega^2 a + \omega a^2 \rangle$.
- where $\omega = e^{2\pi i/3}$. Then the induced representations of the modules are given by
- $\text{Ind}_H^G(W_1) = \mathbb{C}\langle 1_H + a + a^2, b + ba + ba^2 \rangle$.
 - $\text{Ind}_H^G(W_2) = \mathbb{C}\langle 1_H + \omega a + \omega^2 a^2, b + \omega ba + \omega^2 ba^2 \rangle$.
 - $\text{Ind}_H^G(W_3) = \mathbb{C}\langle 1_H + \omega a + \omega^2 a^2, b + \omega ba + \omega^2 ba^2 \rangle$.

Proposition 3.5.7

Let G be a finite group. Let $H \leq G$ be a subgroup. Then there is an adjunction

$$\text{Ind}_H^G : \mathbf{Rep}_H \xrightarrow{-1} \mathbf{Rep}_G : \text{Res}_H^G$$

Theorem 3.5.8 (Frobenius Reciprocity Theorem)

Let G be a finite group. Let $H \leq G$ be a subgroup. Let χ be a character of G . Let ψ be a character of H . Then we have

$$\langle \text{Ind}_H^G \psi, \chi \rangle_G = \langle \psi, \text{Res}_H^G \chi \rangle_H$$

Proposition 3.5.9 (The Mackey Formula)

Let G be a finite group. Let $H \leq G$ be a subgroup. Let ψ be a character of H . Then we have

$$\text{Ind}_H^G \psi(g) = \frac{1}{|H|} \sum_{y \in G, ygy^{-1} \in H} \psi(ygy^{-1})$$

4 Representation Theory of the Symmetric Group

4.1

The conjugacy classes of S_n is the given by the cycle type of the permutations. These are indexed by partitions $\lambda = (\lambda_1, \dots, \lambda_k)$ such that $\sum_{i=1}^k \lambda_i = n$ and $\lambda_i \geq \lambda_{i+1}$. We write $\lambda \vdash n$. These can be drawn into Young diagrams:

- There are λ_i boxes in row i .
- Rows are left justified.

Let S_λ be the subgroup of S_n given by

$$S_\lambda = S_{1, \dots, \lambda_1} \times S_{\lambda_1+1, \dots, \lambda_1+\lambda_2} \times \cdots \times S_{n-\lambda_k+1, \dots, n} \cong S_{\lambda_1} \times \cdots \times S_{\lambda_k}$$

Let M^λ be the module associated to the character $\text{Triv}_{S_\lambda}^{S_n}$. This is not generally irreducible. But we can order partitions of n such that the following are true.

- $M^{\lambda(1)}$ is irreducible (call it $S^{\lambda(1)}$)
- $M^{\lambda(2)}$ contain copies of $S^{\lambda(1)}$ and a unique copy of a new irreducible module $S^{\lambda(2)}$.
- etc...

$S^{\lambda(i)}$ are called Specht modules. $M^{\lambda(i)}$ are called the permutation modules corresponding to λ . The ordering needed is reverse-lexicographic (revlex). Largest number to smallest number, starting with the first component and moving to latter components to break ties.

Let $\bar{t}_i$ denote the distinct λ -tabloids: fill the young diagram of λ with $\{1, \dots, n\}$ (this is called a Young tableau) and mark two as equivalent if each corresponding row has the same elements. Then $M^\lambda = \mathbb{C}\langle \bar{t}_1, \dots, \bar{t}_k \rangle$.

Example 4.1.1

The partitions of $\{1, \dots, 4\}$ is given by

$$\{(4), (3, 1), (2, 2), (2, 1, 1), (1, 1, 1, 1)\}$$

Example 4.1.2

Let $\lambda = (n)$. Then $M^\lambda \cong \mathbb{C}$ and S_n acts on this trivially. Thus $\text{Ind}_{S_\lambda}^{S_n} \text{Triv} = \text{Triv}$.

Let $\lambda = (1, \dots, 1)$. Then $M^\lambda \cong \mathbb{C}^{n!}$. Then $\text{Ind}_{S_\lambda}^{S_n} \text{Triv} = \chi_{\text{reg}}$.

Let $\lambda = (n-1, 1)$. Then $M^\lambda \cong \mathbb{C}^n$. Then $\text{Ind}_{S_\lambda}^{S_n} \text{Triv}$ is the character of the permutation representation.

Definition 4.1.3 (Young Diagram)

Let $n \in \mathbb{N}^+$. Let $\lambda = (\lambda_1, \dots, \lambda_k)$ be a partition of n . Define the Young diagram of λ to be

$$Y(\lambda) = \{(i, j) \in \mathbb{N}^+ \times \mathbb{N}^+ \mid 1 \leq i \leq k, 1 \leq j \leq \lambda_i\}$$

Definition 4.1.4 (Young Tableau)

Let $n \in \mathbb{N}^+$. Let λ be a partition of n . A Young tableau of λ is a bijection $t : Y(\lambda) \rightarrow \{1, \dots, n\}$. Denote the set of all Young tableau of λ by Δ^λ .

Definition 4.1.5 (Young Tabloid)

Let $n \in \mathbb{N}^+$. Let λ be a partition of n .

- Let t_1, t_2 be Young tableau of λ . We say that t_1 and t_2 are equivalent if

$$\{t_1(r_0, j) \mid 1 \leq j \leq \lambda_i\} = \{t_2(r_0, j) \mid 1 \leq j \leq \lambda_k\}$$

for all $1 \leq r_0 \leq k$ (in other words the rows of the tableau are equal up to permutation).

- A Young tabloid of λ is an equivalence class $[t]$ where t is a Young tableau of λ .

Definition 4.1.6 (The Associated Polytabloid of a Tableau)

Let $n \in \mathbb{N}^+$. Let λ be a partition of n . Let t be a λ -tableau. Define the associated polytabloid to be

$$e(t) = \sum_{\sigma \in C(t)} \text{sign}(\sigma)(\sigma \cdot [t])$$

where $C(t) = \prod_{i=1}^k \text{Sym}(C_i)$ where C_i is the i th column of t .

Definition 4.1.7 (The Specht Module of a Partition)

Let $n \in \mathbb{N}^+$. Let λ be a partition for n . Define the Specht module associated to λ to be the $\mathbb{C}$ -vector space

$$S^\lambda = \mathbb{C}\langle e(t) \mid t \text{ is a } \lambda\text{-tableau} \rangle$$

Theorem 4.1.8

Let $n \in \mathbb{N}^+$. Then $\{S^\lambda \mid \lambda \text{ is a partition for } n\}$ forms the complete list of non-isomorphic irreducible $\mathbb{C}[S_n]$ -modules.